

Econometrics

Week 2

Institute of Economic Studies
Faculty of Social Sciences
Charles University in Prague

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Recommended Reading

For today

- Basic regression analysis with time series data
- Chapter 10 (pp. 311 – 342)

For next week

- Further issues in using OLS with time series data
- Chapter 11 (pp. 347 – 370)

Today's Lecture

- Differences between time series data and cross-sectional data.
- Gauss-Markov assumptions for time series regression:
How do they differ from the cross-sectional case?
- Properties of the OLS estimator in time-series context.
- Examples of econometric models using time series data.
- Trends and seasonality.

Cross-Sectional Data vs. Time Series Data

$$y_i = \beta_0 + \beta_1 x_{i1} + u_i \quad u_i \sim N(0, \sigma^2)$$

$$i = 1, 2, \dots, n$$

vs.

$$y_t = \beta_0 + \beta_1 x_{t1} + u_t \quad u_t \sim N(0, \sigma^2)$$

$$t = 1, 2, \dots, T$$

The Nature of Time Series Data

- Time series vs. cross-sections: **Temporal ordering.**
- Until now, we have studied the properties of the OLS estimator based on the assumption that samples are random...
- ...but time series data are not random samples (why?)

Instead, we deal with Stochastic Processes

- “stochastic” from the Greek “stochos”: aim, guess, or characterized by conjecture and randomness.
- The observed data is one realization of a stochastic process.

does it challenge the CLM model?

Classical Linear Model Assumptions

- Assumption MLR.1 (Linear in Parameters)
- Assumption MLR.2 (Random Sampling)
- Assumption MLR.3 (No Perfect Colinearity)
- Assumption MLR.4 (Zero Conditional Mean)
- Assumption MLR.5 (Homoskedasticity)
- Assumption MLR.6 (Normality)

Which of the assumptions is/are challenged when using time-series data?

Classical Linear Model Assumptions

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Which of the assumptions is/are challenged when using time-series data?

Unbiasedness

Repetition from the last lecture

$$\hat{\beta}_{1,OLS} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})} = \dots = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})(u_i)}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + E\left[\frac{\sum_{i=1}^n (x_i - \bar{x})(u_i)}{\sum_{i=1}^n (x_i - \bar{x})^2} | X\right]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} E\left[\sum_{i=1}^n (x_i - \bar{x})(u_i) | X\right]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n E[(x_i - \bar{x})(u_i) | X]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) E[u_i | X]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1$$

Note: All sums go from $t=1$ to $t=T$, i.e. sum over all observations

Unbiasedness in OLS on time series data

- To prove unbiasedness of the OLS estimator, we need that $E[u_i|X] = 0$
- When observations are independent of each other (i.e. under random sampling)
 - $E[u_i|x_i] = 0$ implies that also $E[u_i|X] = 0$
- Without random sampling, we need to explicitly assume the strong form of exogeneity
 - $E[u_i|X] = 0$
 - The expected value of the error term is zero, conditional on all explanatory variables, in the past, contemporaneous, and in the future.

Unbiasedness: Time Series Assumptions

TS1: Linear in parameters

The stochastic process $\{(x_{t1}, x_{t2}, \dots, x_{tk}, y_t) : t = 1, 2, \dots, n\}$ follows the linear model:

$$y_t = \beta_0 + \beta_1 x_{t1} + \dots + \beta_k x_{tk} + u_t, \quad (1)$$

with $\{u_t : t = 1, 2, \dots, T\}$ sequence of error disturbances and T number of observations (time periods).

TS1: Linear in parameters (matrix notation)

The model representing the stochastic process can be written as:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u}, \quad (2)$$

with $\mathbf{y}$ being a $T \times 1$ vector, $\mathbf{X}$ being a $T \times k + 1$ matrix, and $\mathbf{u}$ being a $T \times 1$ vector of error disturbances.

Unbiasedness: Time Series Assumptions

TS2: Zero Conditional Mean

$$E(u_t|X) = 0, \quad t = 1, 2, \dots, T. \quad (3)$$

⇒ error term in any given period is uncorrelated with explanatory variable(s) in all time periods (**past and future!**).

- TS2 implies that explanatory variables are **strictly exogenous**.
- This is needed b/c we don't have random samples.
- A weaker assumption (not assuring unbiasedness in case of time series OLS) is:

$$E(u_t|x_{t1}, \dots, x_{tk}) = E(u_t|x_t) = 0$$

- Means that x_{tj} are **contemporaneously exogenous**.
- Less strict; does not require u_t to be uncorrelated with x_{sj} for $s \neq t$ as in TS2.

Note

In social sciences, many stochastic processes violate the strict exogeneity assumption! Next week we'll learn how to deal with this.

Unbiasedness: Time Series Assumptions

TS3: No Perfect Collinearity

No independent variable is constant or a perfect linear combination of the others.

TS3: No Perfect Collinearity (matrix notation)

The matrix $\mathbf{X}$ has full rank, i.e. it's rank is $k + 1$

These are essentially the same as for cross-sectional data.

Unbiasedness

Theorem 1: Unbiasedness of OLS

Under TS1, TS2, and TS3, the OLS estimators are unbiased conditional on X , and therefore unconditionally as well:

$$E(\hat{\beta}_j) = \beta_j, j = 0, 1, \dots, k.$$

Theorem 1: Unbiasedness of OLS (matrix notation)

Under TS1, TS2, and TS3, the OLS estimator $\hat{\beta}$ is unbiased for β .

- Similar to the cross-sectional case, but we have omitted the random sampling assumption (thanks to assumption TS2)
- Violation of the strict exogeneity assumption brings bias
 - With time series data, **bias occurs even if explanatory variables are contemporaneously exogenous but are NOT strictly exogenous.**
 - With cross-sectional data, **contemporaneous exogeneity is enough for unbiasedness.**

Variance of the OLS estimator

Consider a simple model: $y_t = \beta_0 + \beta_1 \cdot x_t + u_t$

OLS estimator:

$$\hat{\beta}_1^{OLS} = \frac{\sum (x_t - \bar{x})(y_t - \bar{y})}{\sum (x_t - \bar{x})^2} = \beta_1 + \frac{\sum (x_t - \bar{x})u_t}{\sum (x_t - \bar{x})^2}$$

Variance of the OLS estimator (which assumptions are used?):

$$\begin{aligned} Var[\hat{\beta}_1^{OLS} | \mathbf{X}] &= Var\left[\frac{\sum (x_t - \bar{x})u_t}{\sum (x_t - \bar{x})^2} \middle| \mathbf{X}\right] = \frac{Var[\sum (x_t - \bar{x})u_t | \mathbf{X}]}{(\sum (x_t - \bar{x})^2)^2} = \\ &= \frac{\sum (x_t - \bar{x})^2 Var[u_t | \mathbf{X}]}{(\sum (x_t - \bar{x})^2)^2} = \frac{\sigma^2}{\sum (x_t - \bar{x})^2} = \frac{\sigma^2}{SST_x} \end{aligned}$$

Note: All sums go from $t=1$ to $t=T$, i.e. sum over all observations

Variance of the OLS Estimator: Time Series Assumptions

TS4: Homoskedasticity

Conditional on X , the variance of u_t is the same for all t :

$$\text{Var}(u_t|X) = \text{Var}(u_t) = \sigma^2, t = 1, 2, \dots, n.$$

- It means that the error variance should be independent of all x 's (also past and future) and be constant over time.
- When not satisfied, we say that the data are **heteroskedastic**.

Variance of the OLS Estimator: Time Series Assumptions

TS5: No Serial Correlation

Conditional on X , the errors in any two different time periods are uncorrelated:

$$\text{Corr}(u_t, u_s | X) = 0 \quad \text{for all } t \neq s.$$

- It means that the cross-correlations in error (disturbance) structure are independent of all x 's, constant over time and equal to zero.
- When not satisfied, we say that the errors (disturbances) are correlated across time and suffer from **serial correlation** or **autocorrelation**.

We'll talk more about this in two weeks.

Variance of the OLS Estimator: Time Series Assumptions

TS4 and TS5 in matrix notation

Conditional on X , one can write these two assumptions as:

$$\text{Var}(\mathbf{u}|\mathbf{X}) = \sigma^2 \mathbf{I}_T$$

$$\text{Var}(\mathbf{u}) = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1T} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2T} \\ \dots & \dots & \dots & \dots \\ \sigma_{T1} & \sigma_{T2} & \dots & \sigma_{TT} \end{bmatrix} = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix}$$

Variance of the OLS Estimator

Theorem 2: OLS Sampling Variances

Under the time series Gauss-Markov Assumptions (TS1-TS5), the variance of $\hat{\beta}_j$, conditional on X is:

$$\text{Var}(\hat{\beta}_j|X) = \sigma^2/[SST_j(1 - R_j^2)], \quad j = 1, \dots, k,$$

where SST_j is the total sum of squares of x_{tj} and R_j^2 is the R -squared of a regression of x_{tj} on all other explanatory variables.

Theorem 3: Unbiased Estimation of σ^2

Under TS1-TS5, the estimator $\hat{\sigma}^2 = SSR/df$ is an unbiased estimator of σ^2 , where $df = T - k - 1$.

Theorem 4: Gauss-Markov Theorem

Under TS1-TS5, the OLS estimators are the best linear unbiased estimator (BLUE) conditional on X .

Inference Under the Classical Linear Model (CLM) Assumptions

- Under assumptions TS1-TS5, OLS in time-series context has the same desirable finite sample properties as in the cross-sectional data case
- If we add assumption on normality of errors, inference is also the same

TS6: Normality

The errors u_t are independent of X and are independently and identically distributed as $N(0, \sigma^2)$.

TS6 implies TS3-TS5, but is stronger $\rightarrow$ independence and normality.

Inference Under the Classical Linear Model (CLM) Assumptions

Theorem 5: Normal Sampling Distribution

Under TS1-TS6, the CLM assumptions for time series, the OLS estimators are normally distributed, conditional on X . Further, under the null hypothesis, each t statistic has a t distribution, and each F statistic has an F distribution.

Inference Under the Classical Linear Model (CLM) Assumptions

- $\Rightarrow$ everything we have learned about estimation and inference for cross-sectional regressions applies directly to time series regressions.

BUT this also applies to the problems

Inference is only as good as the underlying assumptions!

!!! REMEMBER !!!

- CLM assumptions for time series data are much more restrictive than those for the cross-sectional data.
- In particular, strict exogeneity and no serial correlation.
- $\rightarrow$ unrealistic in social sciences for many data sets.
- We will learn how to overcome this during next lectures

Examples of Time Series econometric models: a static model

A contemporaneous (in Czech: “souběžný”) relation between y and z can be captured by a **static model**:

$$y_t = \beta_0 + \beta_1 z_t + u_t, \quad t = 1, 2, \dots, n.$$

When to use?

- Change in z at time t has immediate effect on y :
 $\Delta y_t = \beta_1 \Delta z_t$, when $\Delta u_t = 0$.
- We would like to know the tradeoff between y and z .

Example: Demand Curve

- Contemporaneous tradeoff between price and consumption:
$$D_t = \beta_0 + \beta_1 \text{Price}_t + u_t$$
- This specification assumes that both parameters are **time-invariant**.

Examples of Time Series econometric models: Finite Distributed Lag (FDL) models

- This is a dynamic model allowing one or more variables to affect y with a **lag**.
- FDL of order two:
$$y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \delta_2 z_{t-2} + u_t,$$
- **FDL model of order q** will include q lags of z .
- Can be used to test the effect of a 1-period temporary increase in z
 - δ_0 – “**impact propensity**” – reflects immediate change in y when z changes by one unit.
 - δ_q – reflects lagged reaction of y to a one unit change in z
- Can be used to test the effect of a permanent increase in z
 - δ_0 – reflects immediate change in y when z changes by one unit.
 - $\delta_0 + \delta_1 + \dots + \delta_q$ – “**long-run propensity**” (LRP)– reflects the long-run change in y after a permanent change in z .

Examples of Time Series econometric models

Note

Static model is a special case of a FDL model

when setting $\delta_0, \delta_1, \dots, \delta_q$ to zero:

$$y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \dots + \delta_q z_{t-q} + u_t$$

$\downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \downarrow$
 $0 \qquad \qquad \qquad 0 \qquad \qquad 0$

Note

Often, we have substantial correlation across z at different lags
 $\Rightarrow$ multicollinearity

Example: Effect of EET (electronic sales record) introduction on prices in restaurants

Since December 2016 restaurants in the Czech Republic were obliged to report each sale through an electronic system. EET was cancelled as of January 1, 2023, but will be reintroduced on January 1, 2027.

- Introduced to reduce tax evasion
- Lot of attention in the **media**
- Opponents worried EET would cause price increase (cf: would cancellation cause price decrease?)
- Use monthly data on restaurant price index (source: Czech Statistical Office) to verify this hypothesis

$$\begin{aligned} priceindex_t = & \alpha_0 + \delta_0 EET_t + \delta_1 EET_{t-1} + \delta_2 EET_{t-2} + \\ & + \delta_3 EET_{t-3} + \delta_4 EET_{t-4} + u_t \end{aligned}$$

Trends in Time Series

- Economic time series often trend (i.e. change regularly over time).
- What happens when we regress two trending time series on each other?

► Example

- We might come across the so called **spurious correlation**
 - We find relationship between two or more trending variables simply because each is growing over time.
 - Think about it as of omitted variable problem:
Trending factors that affect y_t are correlated with an explanatory variable
- It is important to properly account for time trend

Dealing with trending variables

- In presence of spurious correlation, the CLM assumptions are violated
- Which assumption(s) is/are violated?
- How to eliminate this problem?
 - Add a time trend to the regression model, or
 - “Detrend” the trending variables

Adding a time trend to the model

- Simple idea: we add the missing variable to the model
- Example on the board
- Pros:
 - This eliminates the spurious correlation problem
 - It is easy
- Cons:
 - We get unnaturally high R-squared
 - It is impossible to say what portion of total variation in the dependent variable is explained by the explanatory variables (usually we don't think of time trend as of an explanatory variable)

Detrending

- This involves regressing each variable in the model on t ...
- ...and re-running the original model with detrended variables
- Example on the board
- Pros:
 - This eliminates the spurious correlation problem
 - We explicitly see which variables are trending and to what extend
 - R-squared in the final regression is informative of what portion of the variation in the dependent variable is explained by the explanatory variables
- Cons:
 - The procedure is complicated
 - We have to be careful and use the same form of time trend for each variable

Seasonality

- Data observed at monthly or quarterly interval may exhibit *seasonality*, or periodicity.
- Example: Quarterly data on retail sales will tend to jump up in the 4th quarter.
- Seasonality can be dealt with by adding a set of seasonal dummies.
- As with trends, the series can be seasonally adjusted before running the regression.

Thank you for your attention!

Reading for next week

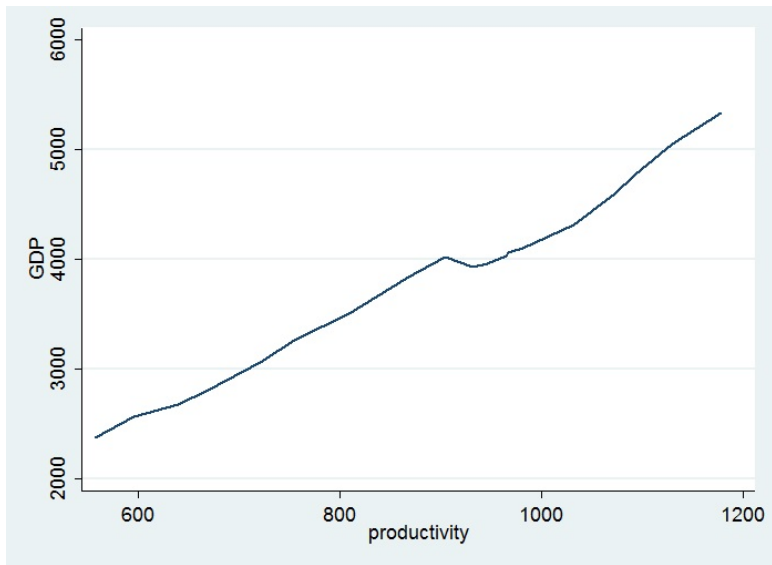
Do not forget to read Chapter 11 for the next week!

Revision for next week

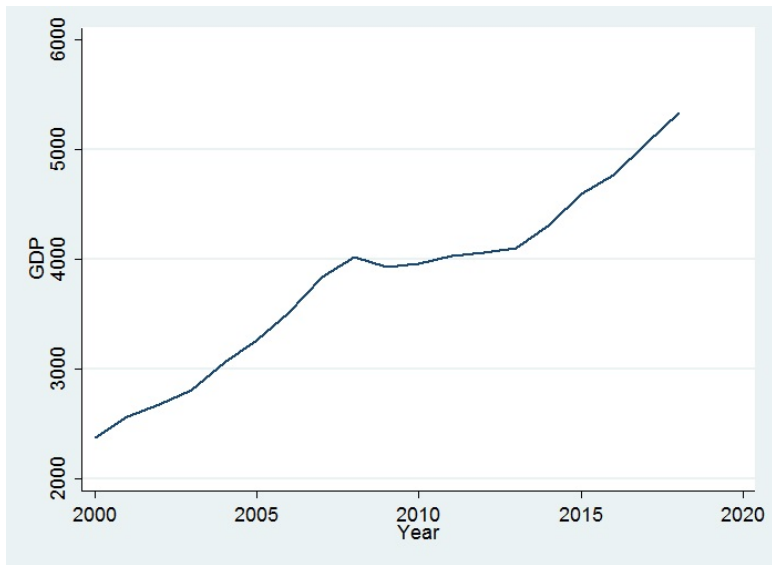
Please, revise also:

- (Statistics) expected value, variance, covariance
- (Econometrics I) asymptotic (large sample) properties of OLS
 - For detailed explanation, watch this nice [explanatory video](#).

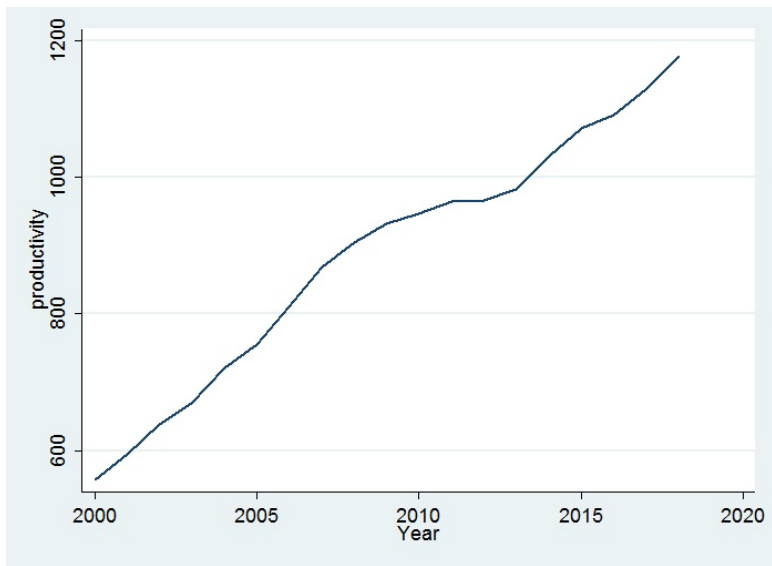
Trends in Time Series



Trends in Time Series



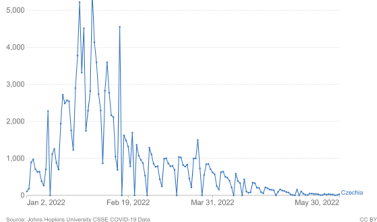
Trends in Time Series



Lagged effects

Daily new confirmed COVID-19 cases per million people

Due to limited testing, the number of confirmed cases is lower than the true number of infections.



Number of COVID-19 patients in hospital

